

Week 9

Chapter 6: Intellectual Property Rights and Computer Technology ¹

Computer Products and Services

Intellectual property rights

Ownership

Intellectual property crimes

Dr. Ali Algarni

1. Ethical and Social Issues in the Information Age 6th edition.

Progress

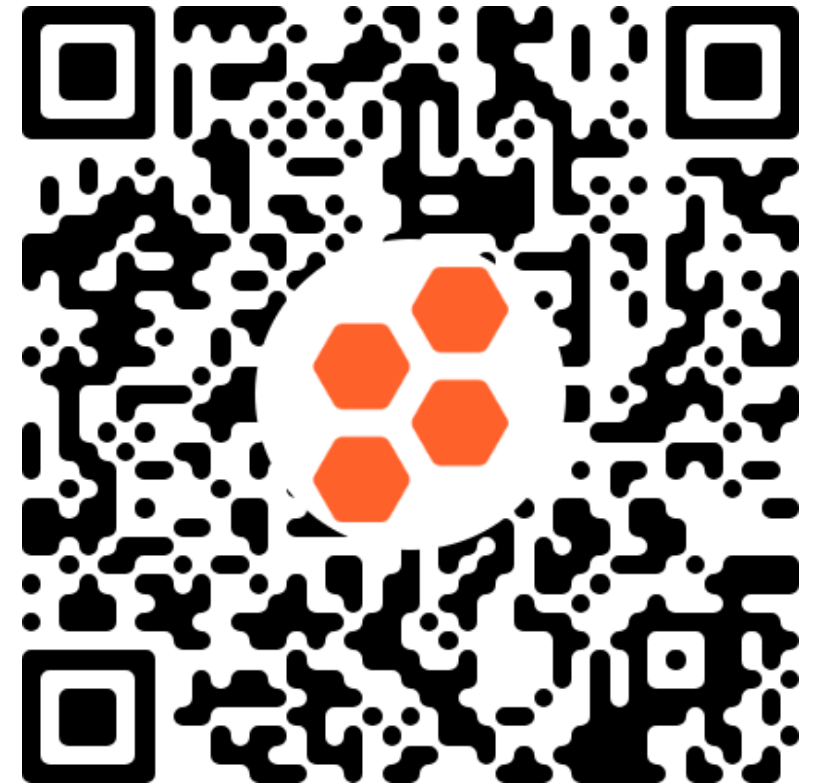
- **Quiz (Socrative)**
- **Assignment**
 - Due to next week before class time
- **next milestones: Project**
 - After Ramadan
 - Short Presentation
- **Project**
 - Papers should be completed
 - Start writing Introduction
 - Start Taxonomy in your paper

Outline

- Intellectual property
- Intellectual property benefits and limits
- Protecting Intellectual property
 - Trade Secret
 - Trademarks/Service marks
 - Patents
 - Copyrights
- Intellectual property in data management

Quiz 2 (Via Socrative)

- Room Number: **CIS6231GROUP1**



Intellectual Property Definition

- Intellectual property: any unique product of the human intellect that has commercial value
 - Books, songs, movies
 - Paintings, drawings
 - Inventions, chemical formulas, computer programs
- **tangible things** that one can claim ownership to.
 - What about intangible ones like data?
- Intellectual property ≠ physical manifestation
- Does right to own property extend to intellectual property?

Some Ethical Questions:

- Government has put many protections for intellectual property; the legal system has given more rights to the creators of intellectual property.
- Are these changes in the best interests of our society, or are politicians catering to special interest groups?
- Do entertainers (ML/AI Developers) have **the right to control** who sees and hears a performance of their music?
- Is it **fair** for some people to **watch shows for free** when others **are paying** to view them?

Intellectual Property benefits

- Some people are **altruistic**; some are not
- Allure of wealth can be an incentive for speculative work
 - Has the right to benefit
- Authors of U.S. Constitution recognized benefits to limited intellectual property protection
- Promotion of invention Culture
- Dissemination of technical information
- Collateral to obtain financing

Intellectual Property Limits

- Giving creators rights to their inventions stimulates creativity
- However, Society benefits most when inventions in public domain
- Congress has struck compromise by giving authors and E.g., rights for a limited time
 - E.g. patent
 - Open-source Software

Protecting of Intellectual Property

- Four types:
 - Trade secrets
 - Trademarks and service marks
 - Patents
 - Copyrights
- Each one is discussed using three criteria:
 - Public domain
 - Duration
 - Applications

Trade Secret

- Confidential piece of intellectual property that gives company a competitive advantage
- Never expires
- Not appropriate for all intellectual properties
- Reverse engineering allowed
- May be compromised when employees leave firm
- Society Effect
 - In US, no federal protection, it state law
- Examples:
 - Secret formula for Coca-Cola
 - Google's search algorithm



Trade Secret Characteristics

- The extent to which the information is known outside the business
- The extent of measures taken by individuals possessing the trade secret to guard the secrecy of the information
- The value of the information to the owner and to the competitor
- The amount of effort or money spent by the owner to develop or gather the information.
- The ease or difficulty with which the information could be properly acquired or duplicated by others

Trademark, Service Mark

- Trademark: Identifies goods
- Service mark: Identifies services
 - **American Airlines®**.an American company that provides flight services.
- Company can establish a “brand name”
- Does not expire
- If brand name becomes common noun, trademark may be lost
- Companies advertise to protect their trademarks
- Companies also protect trademarks by contacting those who misuse them



Patent

- A public document that provides detailed description of invention
- Provides owner with exclusive right to the invention
- Owner can prevent others from making, using, or selling invention for 20 years
- Three Types:
 - Utility
 - Design
 - Plant

Patent



US006763791B2

(12) **United States Patent**
Gardner et al.

(10) **Patent No.:** **US 6,763,791 B2**
(45) **Date of Patent:** **Jul. 20, 2004**

(54) **CAM PHASER FOR ENGINES HAVING TWO CHECK VALVES IN ROTOR BETWEEN CHAMBERS AND SPOOL VALVE**

(75) Inventors: **Marty Gardner**, Ithaca, NY (US);
Michael Duffield, Medina, NY (US)

(73) Assignee: **BorgWarner Inc.**, Auburn Hills, MI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **10/198,476**

(22) Filed: **Jul. 18, 2002**

(65) **Prior Publication Data**

US 2003/0033999 A1 Feb. 20, 2003

Related U.S. Application Data

(60) Provisional application No. 60/312,140, filed on Aug. 14, 2001.

(51) Int. Cl.⁷ **F01L 1/34**

(52) U.S. Cl. **123/90.17; 125/90.15**

(58) Field of Search **123/90.13, 90.15**

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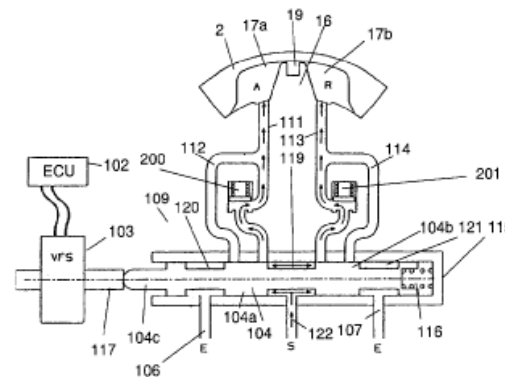
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Greg Dziegielewski

(57) ABSTRACT

An infinitely variable camshaft timing device (phaser) has a control valve located in the rotor. Since the control valve is in the rotor, the camshaft need only provide a single passage for supplying engine oil or hydraulic fluid, and does not need multiple passageways for controlling the phaser, as in the prior art. Two check valves, an advance chamber check valve and a retard chamber check valve, are also located in the rotor. The check valves are located in the control passages for each chamber. The main advantage of putting the check valves in the advance and retard chambers instead of having a single check valve in the supply is to reduce leakage. This design also eliminates high pressure oil flow across the spool valve and improves the response time of the check valve to the torque reversals due to a shorter oil path. In addition, the phaser of the present invention outperforms an oil pressure actuated device and consumes less oil.

14 Claims, 6 Drawing Sheets



Software Patents

- Inventions can be patented, **but not algorithms**
 - 3D Face Recognition Scanner vs CNN?
 - Federal patent law states that mathematical formulas may not be patented
- Patent Office having a hard time determining prior art
- Result: some bad patents have been issued
- General skepticism about value of software patents

Licensing Patents

- **Microsoft** invented a software technology called ClearType that improves the clarity of text in LCD/LED monitors.
- It is protected by 10 patents.
- In 2003 Microsoft started to license it to other companies with a certain **royalty rate**.



Royalty

[rɔɪ(-ə)l-tē]

A payment made to an individual or company for the ongoing use of their assets, including copyrighted works, franchises, and natural resources.

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Software Copyrights

- Copyright protection began 1964
- What gets copyrighted?
 - Expression of idea, not idea itself
 - Object program, not source program
- Companies treat source code as a trade secret
- Copyright Violations
 - Copying a program to give or sell to someone else
 - Preloading a program onto the hard disk of a computer being sold
 - Distributing a program over the Internet

Software Copyrights

- Provides owner of an original work five rights
 - Reproduction
 - Distribution
 - Public display
 - Public performance (museum/ public AI exhibit)
 - Production of derivative works
- Copyright-related industries represent 6% of U.S. gross domestic product (> \$900 billion/yr)
- Copyright protection has expanded greatly since 1970

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Open-Source Software

- No restrictions preventing others from selling or giving away software
- Source code included in distribution
- No restrictions preventing others from modifying source code
- No restrictions regarding how people can use software
- Same rights apply to everyone receiving redistributions of the software

Open-Source Software Benefits

- Gives everyone opportunity to improve program
- New versions of programs appear more frequently
- Eliminates tension between obeying law and helping others
- Programs belong to entire community
- Shifts focus from manufacturing to service

Open-Source Software Issues

- Without critical mass of developers, quality can be poor
- Without an “owner,” incompatible versions may arise
- Relatively weak graphical user interface
- Poor mechanism for stimulating innovation (no companies will spend billions on new programs)

Introduction to intellectual property rights in data management(Cornell Data Service/Univ.)

- A builder of a database or other data resource (e.g. datasets) will have an interest in who owns that resource and how others may use it.
- **Data vs Databases**
 - Data can't be copyrighted or licensed vs collected and organized data
- It is important to understand data licensing from the perspective of both the data user and data creator.

Data licensing

- it is imperative that others know the terms of use for the database and the data content.
- the [Open Data Commons](#) group has been developing legally binding tools to govern the use of data sets.
- The three ODC licenses are:
 - [Public Domain Dedication and License](#) (PDDL): This dedicates the database and its content to the public domain, free for everyone to use as they see fit.
 - [Attribution License](#) (ODC-By): Users are free to use the database and its content in new and different ways, provided they provide attribution to the source of the data and/or the database.
 - [Open Database License](#) (ODC-ODbL): any subsequent use of the database must provide attribution, an unrestricted version of the new product must always be accessible and distributed using the same terms. It is the most restrictive of all ODC licenses.
- Kristin Briney Copyright and Research data: <https://widgets.figshare.com/articles/3117838/embed>

Creative Commons Licenses²

- It has a library of standardized licenses, and some of them apply to data and databases.
- Two CC licenses that are of greatest relevance to data management are:
 - [CC0](#) (i.e., “CC Zero”): When an owner wishes to waive her copyright and/or database rights, she can use the CC0 mark. It effectively places the database and data into the public domain. It is the functional equivalent of an ODC PDDL license.
 - [Public Domain mark](#) (PDM): It is used to mark works that are in the public domain, and for which there are no known copyright or database restrictions. It is possible to flag factual data as PDM in a database, for example, in order to make it clear it is free to use.

Creative Commons Licenses²

- Example: Heart Disease dataset in UCI ML repositories
 - <https://archive.ics.uci.edu/dataset/45/heart+disease>
 - licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

Sharing data that you have collected from other sources³

- Licensed data can have restrictions in the way it can be used or shared downstream
- When re-using existing data be sure to clarify ownership, obtain permissions if needed, and understand limits set by licenses
- Be sure to provide appropriate attribution and citation
- If licensing restricts sharing of the data, providing detailed information about where the data were obtained and how the data were analyzed can help with reproducibility

3. Intellectual Property: Harvard Longwood Research data management . <https://datamanagement.hms.harvard.edu/share-publish/intellectual-property>

Outline

- 4 weeks left
 - Data, media and Cybersecurity issues in
 - Governance and Data management legal and ethical issues
- Project 2 submission is due next week (Thursday)
 - Survey paper (taxonomic paper [here](#))
- Assignment 2
- Today's Class
 - IP four forms: Trade secret, trademarks, patent, copyright
 - IP ownership and Crime
 - Protecting Ownership Right

Intellectual Property Ownership

- An idea is novel if it is original, authentic, and new.
 - Inventiveness, creativity, and discoveries
- Before an idea can be useful, however, it must be put into utilizable form, either as a process or as an application.
- That is the core of **intellectual property rights**.
- In many countries, the owner of such an application for the idea has a set of legal **rights** to its expression as a creation, work, invention, discovery, information, or any other form.
- These **rights** are grouped into subsets defining specific areas of interests such as the right to make, use, or sell one's works of discovery or creation.

The Politics of Ownership

- lot has been written about the concept of intellectual property rights
 - US government negotiated with China, West believe has the highest rate of abuse of the intellectual property laws.
- Many statements made about the effects of **cultural** differences
- West continues to keep the lead in technological development, other culture will be changed
 - Already a number of countries in Southeast Asia have been forced to abide by West laws
- Another issue is perception controversy.
- Many people believe the protection of the manifestation of one's ideas by copyrights, patents, trademarks, and trade secrets laws automatically constitutes a monopoly of the benefits that come with the ideas.
- the US Congress and other governments have passed antitrust laws to calm the public.

The Psychology of Ownership

- we are aware of the distinct psychology about individual ownership.
- We tend to classify those items that we individually own, whether they are few or abundant, according to their intrinsic value to us.
- We may believe our self-worth and status in the community depend on the intrinsic value of the items we own. So the intrinsic value we attach to these items is one of the most important aspects of ownership.
- Another aspect of ownership is the **tangibility of what we own**.
- it tends to raise status in the community, gain the respect of others
 - turn affects our self-esteem and our egos.

Intellectual Property Crimes

- An intellectual property crime (IPC) is the act of infringe on the rights of the owners of the intellectual property.
- IPC refers to all activities that involve **infringement**, **counterfeiting**, and **piracy** of products and services for profit without.
- Technological advances have made these crimes grow like a wildfire in the past decade
 - committing these crimes a lot easier.
- Technology increased these crimes by making the costs of duplicated products low and easy to make.

Intellectual Property Crimes- Infringement

- This legal protection of domain of rights is **offered in subsets of laws** that define the boundaries that can be enforced.
- Anybody else with no rights is considered an infringer.
- There are three types of infringements:
 - **Direct infringement:** The infringer knowingly or otherwise makes, uses, sells, or copies a protected item without any alteration.
 - **Inducement infringement:** intentionally supports infringement activities without individual involved.
 - **Contributory infringement:** The infringer takes part in the infringement of a protected item.

Intellectual Property Crimes- Infringement

- Copyright Infringement
 - Copyright infringement is very difficult to prove
 - However, **US courts** have provided guidelines that many courts normally follow
 - Whether the infringer has knowledge or visual contact with the work
 - Whether the work under dispute is a major revision with substantially new contents of the original or just a variation
 - Example: Article 21 is Saudi Authority for Intellectual Property (SAIP):
<https://externalportal-backend-production.saip.gov.sa/sites/default/files/2022-06/copyright%20law%201.pdf>

Intellectual Property Crimes- Infringement

- **Trademark Infringement**

- Trademark infringement is very difficult to prove
- If proved, infringer may be asked to pay any or a combination of the following:
 - monetary awards, losses and damage of infringement, and legal fees.

- **Patent Infringement**

- Highly sophisticated methods of policing and investigative work need to be followed.
- public law enforcement can't be used
- patent owners tend to use a device that uses the public to do the policing for them. They achieve this by using patent markings on their products,

Protection of Ownership Rights

- **Domains of protections**

- *Copyrights*: protect all rights embodied within the copyrighted work by the copyright act of the particular country, including the right to use, transform, sale, copy, and modify.
- *Patents*: Patent laws protect all rights embodied in the particular country's patent law.
- *Trademarks*: Trademark laws protect all rights in the different trademark statutes depending on the state and country.
- *Trade secrets*: Trade secret statutes and laws protect all rights within the different states, local authority, and country's statutes.

- **Source and Type of Protection**

- USA, by the copyright and patent laws

Protection of Ownership Rights

- **Source and Type of Protection**

- USA:

- The Antipiracy Act of 1976
 - The Communication Act of 1984 The No Electronic Theft Act (NET Act)
 - The Digital Millennium Copyright Act (DMCA)
 - The Economic Espionage Act of 1996
 - Money Laundering Act of 1956

- Global:

- The World Trade Organization (WTO)
 - Interpol
 - The Universal Copyright Convention (UCC)
 - The Trade-Related Aspects of Intellectual Property Rights (TRIPPS)

- Saudi? **Class Activity**

Protection of Ownership Rights

- **Duration of Protection**

- Depends on a number of factors, including the body of laws
- Protecting your rights and your **geographical region**.

- **Strategies of Protection**

- Duty of individuals to devise strategies to safeguard these IP rights
- Methods have been used by individuals and companies to defend themselves.
 - Spying on competitors and suspected infringers using private undercover operatives
 - Collaborating with government officials to check on imports and exports.
 - Call governments when suspected foreign infringements.

Protecting Computer Software Under the IP

- Software Piracy
 - software piracy as the act of copying, distributing, or using proprietary software.
 - It has been illegal ever since software started being protected by law after software manufacturers started filing for patents and copyrights
 - It is complex issue: using illegal software without knowing
- Protection of Software Under Copyright Laws
 - Section 101 of the 1980 US Copyright Amendment: protect software and its documentation
 - Buyer has also right to make more copies (backup) in same machine
- Protection of Software Under Patent Laws
- Protection of Software Under Trademarks

Protecting Computer Software Under the IP

- Protection of Software Under Patent Laws
 - Unlike the copyright laws, patents protect the processing and implementation of the idea
 - Software also may be protected under the patent laws under certain circumstances.
 - Issues includes software many not be patentable, patent registration expenses for one-person software and total disclosure of all information
- Protection of Software Under Trademarks
 - Trademarks prevent information disclosure
 - Software development start with blueprint and flowchart, then coding
 - blueprint and flowchart not be known outside design circles.
 - If blueprint and flowchart was known, then easy to develop.
 - Case: what if employee not yet developer was resigned, and then after years become developer and use the flowchart for new software. How could be protected
 - Nondisclosure contracts? How many years